Payment-Enabled Agents (ACP, AP2, x402, MPP)

Mode 2 Phase 3 (Scale the Workforce), Chapter 9/9 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/payment-enabled-agents-crash-course

EN: Mode 2's final chapter. 4 payment protocols are LAYERS, not rivals: Discovery, Authorization, Commerce, Settlement. Read the use case, split it into the 4 layers, stack the right protocol per layer.
Roman Urdu: Mode 2 ka aakhri chapter. 4 payment protocols LAYERS hain, rivals nahi: Discovery, Authorization, Commerce, Settlement. Use case parho, 4 layers mein todo, har layer par sahi protocol stack karo.

1. Why New Protocols + 4 Layers

- Payment systems assume a human clicks 'buy' -- breaks for agents: no email/account, high-frequency looks like an attack, no phone for disputes.
- **4 layers, 4 different incumbents:** Discovery (MCP/A2A/directories) -> Authorization (AP2/ACP SPT/TAP/ERC-8004) -> Commerce (ACP/UCP) -> Settlement (x402/MPP/card rails).
- **Core rule:** one protocol per layer, compose across layers.

2. The Four Protocols

- **ACP** (OpenAI+Stripe) -- consumer checkout, one-time scoped SPT token, merchant is Merchant of Record. **AP2** (Google+60 partners) -- never moves money itself, only PROVES via a mandate chain (Intent->Cart->Payment), non-repudiable audit trail.
- **x402** (Coinbase->Linux Foundation) -- HTTP 402 revived, stablecoin, 1-2 sec, no account/human -- but NO formal dispute mechanism (its biggest limit). **MPP** (Stripe+Tempo) -- prepaid-session settlement, multi-rail.

3. Composition Rules + Spend Limits

- Test for composing vs competing: are two protocols on the SAME layer (compete) or different layers (compose)? Common examples: AP2+ACP, AP2+x402, ACP+x402.
- **Decimal, never float, for money** in the Pydantic contract layer. tool_input_guardrail blocks payment BEFORE it happens -- output_guardrail fires too late, after payment already occurred.
- **3 independent spend-limit levels:** wallet/payment-method limits (real protection, holds even if agent code fails), SDK guardrail (per-call, pre-payment), application business logic. Trap: protocol caps don't add across protocols (100x $50-SPT = $5,000).

4. Identity + Disputes

- The private signing key IS the identity -- never a wallet address, keep it in a vault, never in env vars. Rotate keys on a 90-day baseline.
- Dispute strength by protocol: ACP (card-network chargeback, strong), AP2 (mandate chain as evidence), MPP (inherits Stripe's dispute machinery), x402 (no formal dispute -- its biggest limitation).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
4 Layers	Discovery, Authorization, Commerce, Settlement.	Discovery, Authorization, Commerce, Settlement.
One protocol per layer, compose across layers	The core composition rule -- never treat same-layer protocols as compatible add-ons.	Core composition rule -- same-layer protocols ko compatible add-ons mat samjho.
tool_input_guardrail	Blocks a payment call BEFORE it happens -- the course's payment-safety spine.	Payment call hone SE PEHLE block karta -- course ka payment-safety spine.
Mandate (AP2)	A signed, non-repudiable proof chain (Intent->Cart->Payment) -- doesn't move money itself.	Signed, non-repudiable proof chain (Intent->Cart->Payment) -- khud paisa move nahi karta.
3 independent spend-limit levels	Wallet limits, SDK guardrail, application business logic -- caps don't stack automatically.	Wallet limits, SDK guardrail, application business logic -- caps automatically stack nahi hote.

Quick Self-Check

Question	Answer
Why can't one payment protocol win across all 4 layers?	Each layer has a different incumbent with its own leverage (search, OAuth, Stripe/Shopify, Visa/crypto) -- new protocols each took the one layer where their sponsor had leverage.
Why must tool_input_guardrail be used instead of output_guardrail for payments?	output_guardrail fires too late -- after the payment has already happened; tool_input_guardrail blocks it BEFORE it occurs.
Why doesn't relying only on protocol-level caps protect against runaway spend?	Caps don't add across protocols or calls -- 100 separate $50 ACP tokens can still total $5,000, so independent wallet and application-level limits are also needed.
Why is x402's lack of a formal dispute mechanism its biggest limitation?	Stablecoin settlement is fast and final with no built-in chargeback path -- any dispute protection has to be bolted on separately (escrow, AP2 composition, seller guarantees).

Where This Leads / Sources

Completes Mode 2 (Manufacturing). Full Decision Lab (5 worked examples incl. a non-Stripe/non-OpenAI rebuild) in Part 5; Production Checklist + Design-Review Template in Part 7.

Source: docs/payment-enabled-agents-crash-course/ (this repo), content originally via Zia Tutor AI connector.